EHMIS Tokha · DHIS2

CVD Risk Assessment Tool

User Guide for Clinical Staff

A step-by-step guide to using the automatic 10-year cardiovascular disease (CVD) risk calculator built into the Out Patient Service form in Capture.

Important: This tool produces a risk *estimate*, not a diagnosis. It does not prescribe treatment. Clinical decisions must follow your facility's approved protocols.

Version 0.1.0 · EHMIS Tokha (erecord.hmis.gov.np/tokha) · Program: बहिरङ्ग सेवा (OPD)

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1. What this tool does

When you open an Out Patient Service visit for a patient in Capture, a panel titled **"10 years CVD Risk"** appears on the right-hand side of the screen. It automatically:

- Reads the patient's age and sex from their record.
- Reads smoking status, blood pressure, cholesterol, and diabetes status as you fill them into the visit form.
- Calculates the patient's 10-year cardiovascular disease risk using the official **WHO 2019 risk chart (South Asia region)**.
- Displays the result immediately, color-coded, as you type — no extra data entry, no separate form, and no button to click to calculate.

> **You don't do anything extra.** Fill in the visit form as you normally would. The risk score appears and updates by itself.

2. Where to find it

1. Open the patient's enrollment in Capture, as usual.
2. Click **New event** (or open an existing visit you're editing) under **Out Patient Service**.
3. Look to the right of the form. The **"10 years CVD Risk"** panel is there from the moment the page loads.

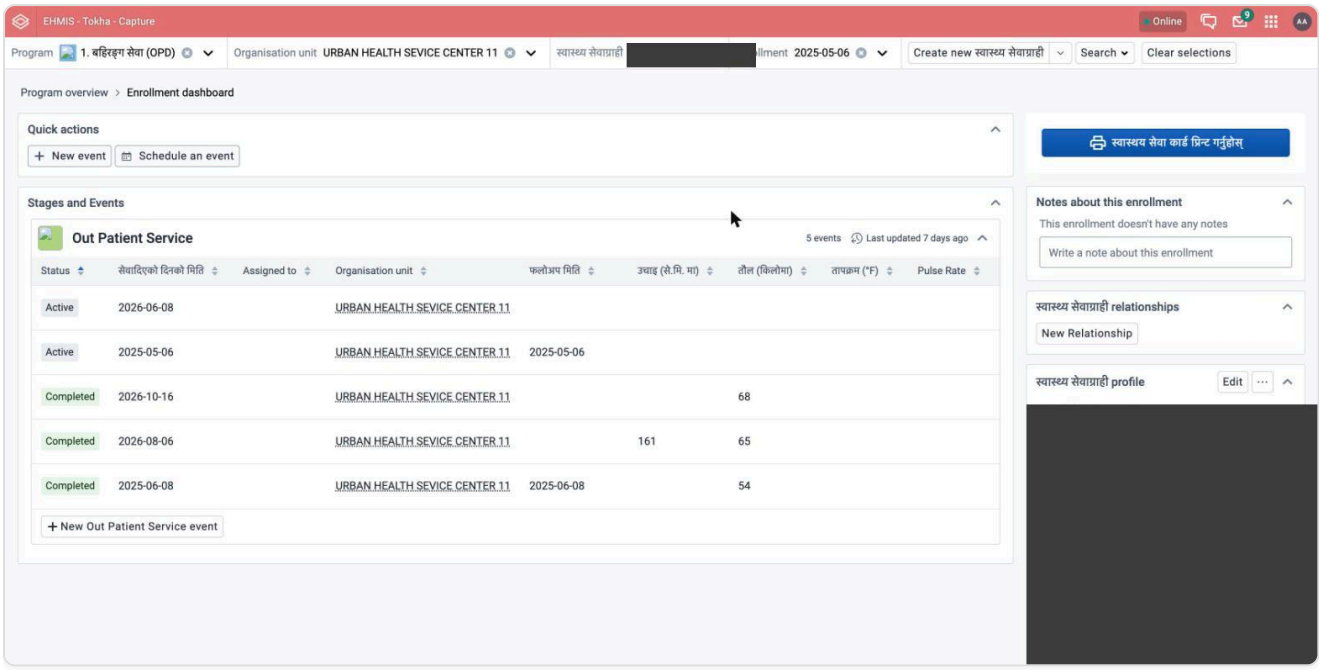


Figure 1 — The enrollment dashboard. Click "New event" to start a visit.

3. Step-by-step: getting a risk score

When a new visit opens, the panel starts empty — every row shows a dash (-) because nothing has been entered yet.

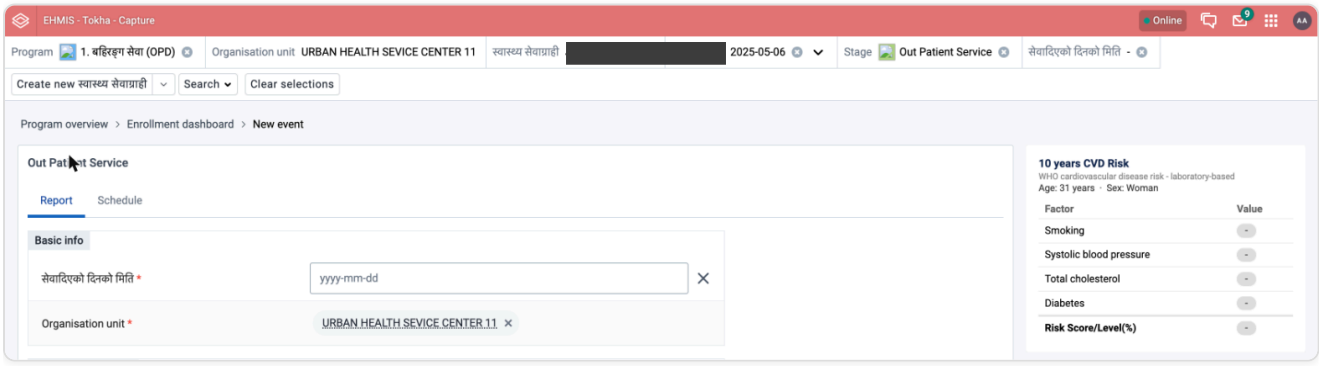


Figure 2 — A new visit, before any clinical data has been entered. Every row shows "-".

1. Enter the visit date as usual.

2. Tick **"मेयर दिर्घायु कार्यक्रम"** (or select the NCD service type) to reveal the clinical fields: smoking, blood pressure, cholesterol, and diabetes.

3. Fill in each field exactly as you normally would:
   - **Smoking status** (धुम्रपान)
   - **Systolic blood pressure** (रक्तचाप)
   - **Total cholesterol**
   - **Diabetes status** (मधुमेह)

4. As each field is filled in, watch the panel on the right — it updates **automatically**, row by row, with no action needed from you.

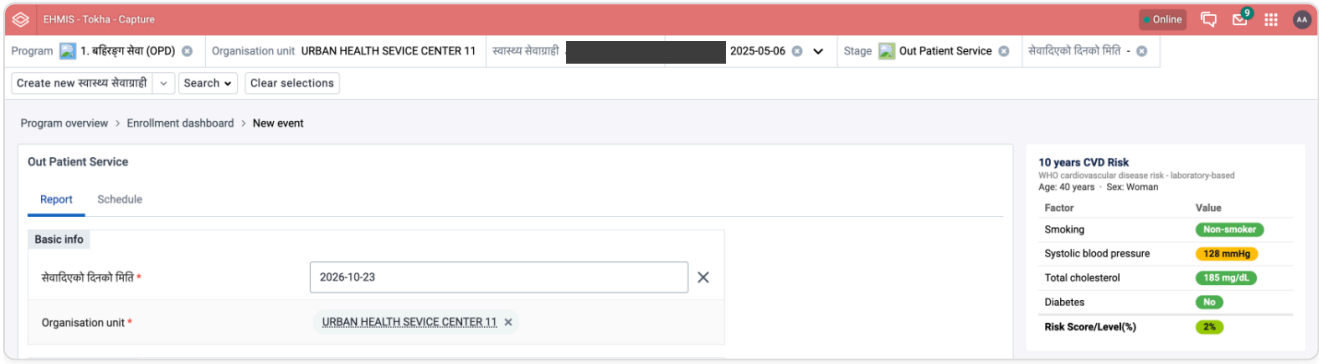


Figure 3 — Example: all values entered. Every factor is color-coded, and the overall 10-year risk (2%) is shown at the bottom.

> The result updates instantly if you go back and change a value — for example, correcting a blood pressure reading immediately recalculates the risk score.

4. Understanding the result table

Each row shows one factor, its value, and a color badge showing whether it's in a normal, borderline, or high range:

Badge color	Meaning
Normal	The value is within the normal clinical range.
Borderline	The value is elevated — worth noting, not yet high-risk on its own.
High	The value is outside the normal range and contributes to higher risk.
-	Not entered yet, or could not be found on this visit.

The final row, **Risk Score/Level(%)**, is the overall 10-year cardiovascular risk estimate. Its color follows the official WHO chart's own legend:

Risk band	Color
Under 5%	<5%
5% to under 10%	5–10%
10% to under 20%	10–20%
20% to under 30%	20–30%
30% or more	≥30%

An example with several factors elevated, showing how the panel looks for a higher-risk patient:

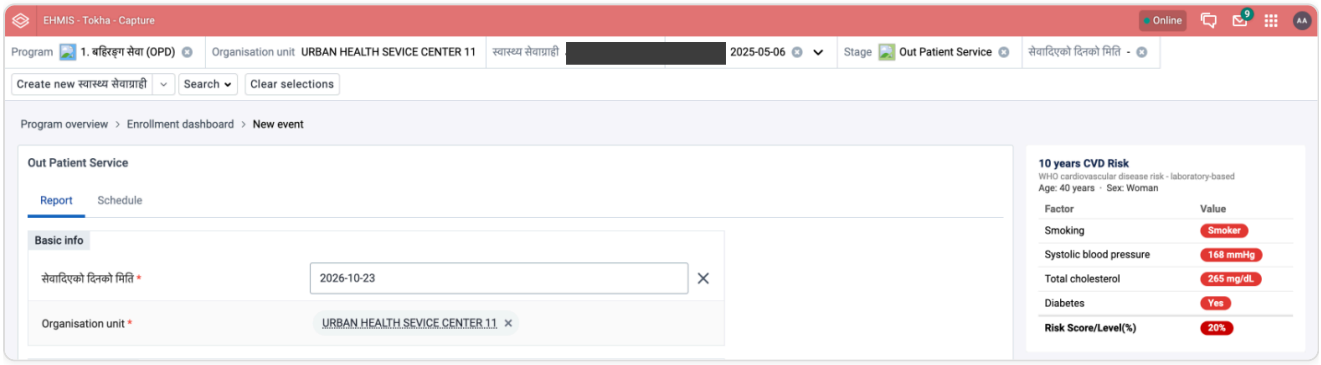


Figure 4 — Example: several factors elevated. The overall risk (20%) is shown in the matching WHO chart color.

5. Saving the result

> **There is nothing extra to save.** The calculated score is written automatically into the visit's own Risk Score field. Simply click your usual **Save**, **Complete**, or **Update** button at the bottom of the form, exactly as you always do.

There is no separate "Save risk" button and no separate risk-assessment form — the score travels with the rest of the visit's data.

6. What if a value is missing?

If a factor hasn't been entered, its row simply shows a dash (-). The tool never guesses or fills in a value on its own.

- If **total cholesterol** and **diabetes status** are both entered, the official WHO laboratory-based chart is used.

- If either is missing but **height and weight** have been entered, the tool automatically falls back to the WHO's alternative BMI-based chart instead — this is a documented WHO method, not an approximation.
- If neither is available, the Risk Score row stays as "-" until enough information is entered.

7. Frequently asked questions

Do I need to open a separate app?

No. The panel is built directly into the Out Patient Service visit page in Capture.

Can I edit the risk score myself?

No. It is calculated automatically and is read-only — this ensures every score is calculated the same, consistent way.

Why does the Age shown change?

Age is calculated from the patient's date of birth as of the visit date you entered — not always "today's" age. This keeps old, already-completed visits showing the patient's age *at that visit*.

What if I'm reviewing an old, already-saved visit?

Open it in **Edit event** mode as usual. The panel reads the saved values and displays the risk exactly as it would for a new visit, and will update live if you change a value.

Is this a diagnosis?

No. It is a risk estimate based on the WHO 2019 chart. Clinical judgement and your facility's protocols always take priority.